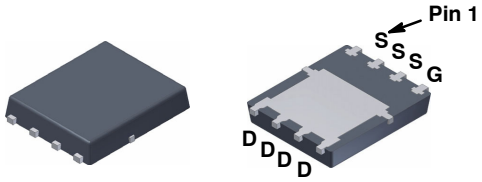
# MOSFET - P-Channel, POWERTRENCH®

## -30 V, -122 A, 3.2 mΩ

# FDMS6681Z




Top Bottom

Power 56 (PQFN8)
CASE 483AE

### General Description

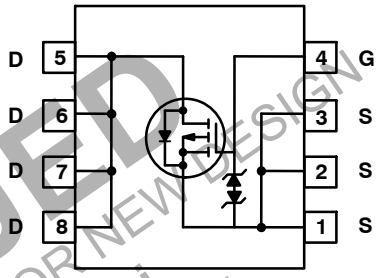
The FDMS6681Z has been designed to minimize losses in load switch applications. Advancements in both silicon and package technologies have been combined to offer the lowest $R_{DS(on)}$ and ESD protection.

### Features

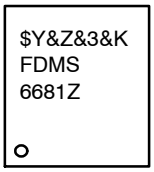
- Max $R_{DS(on)}$ = 3.2 mΩ at $V_{GS}$ = −10 V, $I_D$ = −21.1 A
- Max $R_{DS(on)}$ = 5.0 mΩ at $V_{GS}$ = −4.5 V, $I_D$ = −15.7 A
- Advanced Package and Silicon Combination for Low $R_{DS(on)}$
- HBM ESD Protection Level of 8 kV Typical (Note 3)
- MSL1 Robust Package Design
- RoHS Compliant

### Applications

- Load Switch in Notebook and Server
- Notebook Battery Pack Power Management




### MARKING DIAGRAM




| | |
|---|---|
| $Y | = **onsemi** Logo |
| &Z | = Assembly Plant Code |
| &3 | = Numeric Date Code |
| &K | = Lot Code |
| FDMS6681Z | = Specific Device Code |

### ORDERING INFORMATION

See detailed ordering and shipping information on page 6 of this data sheet.

### MOSFET MAXIMUM RATINGS ($T_A$ = 25°C unless otherwise noted)

| Symbol | Parameter | Ratings | Unit |
|---|---|---|---|
| $V_{DS}$ | Drain to Source Voltage | −30 | V |
| $V_{GS}$ | Gate to Source Voltage | ±25 | V |
| $I_D$ | Drain Current − Continuous $T_C$ = 25°C (Note 5) | −122 | A |
| | − Continuous $T_C$ = 100°C (Note 5) | −77 | |
| | − Continuous $T_A$ = 25°C (Note 1a) | −21.1 | |
| | − Pulsed (Note 4) | −600 | |
| $P_D$ | Power dissipation $T_C$ = 25°C | 73 | W |
| | Power dissipation $T_A$ = 25°C (Note 1a) | 2.5 | |
| $T_J$, $T_{STG}$ | Operating and Storage Junction Temperature Range | −55 to +150 | °C |

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

### THERMAL CHARACTERISTICS

| Symbol | Parameter | Ratings | Unit |
|---|---|---|---|
| $R_{\theta JC}$ | Thermal Resistance, Junction to Case | 1.7 | °C/W |
| $R_{\theta JA}$ | Thermal Resistance, Junction to Ambient (Note 1a) | 50 | |

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**ELECTRICAL CHARACTERISTICS** ($T_J$ = 25°C unless otherwise noted)

| Symbol | Parameter | Test Conditions | Min | Typ | Max | Unit |
|---|---|---|---|---|---|---|
| **OFF CHARACTERISTICS** | | | | | | |
| $BV_{DSS}$ | Drain to Source Breakdown Voltage | $I_D$ = −250 μA, $V_{GS}$ = 0 V | −30 | – | – | V |
| $\frac{\Delta BV_{DSS}}{\Delta T_J}$ | Breakdown Voltage Temperature Coefficient | $I_D$ = −250 μA, referenced to 25°C | – | 20 | – | mV/°C |
| $I_{DSS}$ | Zero Gate Voltage Drain Current | $V_{DS}$ = −24 V, $V_{GS}$ = 0 V | – | – | −1 | μA |
| $I_{GSS}$ | Gate to Source Leakage Current | $V_{GS}$ = ±25 V, $V_{DS}$ = 0 V | – | – | ±10 | μA |
| **ON CHARACTERISTICS** | | | | | | |
| $V_{GS(th)}$ | Gate to Source Threshold Voltage | $V_{GS}$ = $V_{DS}$, $I_D$ = −250 μA | −1 | −1.7 | −3 | V |
| $\frac{\Delta V_{GS(th)}}{\Delta T_J}$ | Gate to Source Threshold Voltage Temperature Coefficient | $I_D$ = −250 μA, referenced to 25°C | – | −7 | – | mV/°C |
| $R_{DS(on)}$ | Static Drain to Source On Resistance | $V_{GS}$ = −10 V, $I_D$ = −22.1 A | – | 2.7 | 3.2 | mΩ |
| | | $V_{GS}$ = −4.5 V, $I_D$ = −15.7 A | – | 4.0 | 5.0 | |
| | | $V_{GS}$ = −10 V, $I_D$ = −22.1 A, $T_J$ = 125°C | – | 3.9 | 5.0 | |
| $g_{FS}$ | Forward Transconductance | $V_{DD}$ = −10 V, $I_D$ = −22.1 A | – | 143 | – | S |
| **DYNAMIC CHARACTERISTICS** | | | | | | |
| $C_{iss}$ | Input Capacitance | $V_{DS}$ = −15 V, $V_{GS}$ = 0 V, f = 1MHz | – | 7803 | 10380 | pF |
| $C_{oss}$ | Output Capacitance | | – | 1540 | 2050 | |
| $C_{rss}$ | Reverse Transfer Capacitance | | – | 1345 | 2020 | |
| **SWITCHING CHARACTERISTICS** | | | | | | |
| $t_{d(on)}$ | Turn – On Delay Time | $V_{DD}$ = −15 V, $I_D$ = −22.1 A, $V_{GS}$ = −10 V, $R_{GEN}$ = 6 Ω | – | 15 | 24 | ns |
| $t_r$ | Rise Time | | – | 38 | 61 | |
| $t_{d(off)}$ | Turn – Off Delay Time | | – | 260 | 416 | |
| $t_f$ | Fall Time | | – | 197 | 316 | |
| $Q_g$ | Total Gate Charge | $V_{GS}$ = 0 V to −10 V; $V_{DD}$ = −15 V, $i_D$ = −22.1 A | – | 172 | 241 | nC |
| $Q_g$ | Total Gate Charge | $V_{GS}$ = 0 V to −5 V; $V_{DD}$ = −15 V, $i_D$ = −22.1 A | – | 97 | 136 | |
| $Q_{gs}$ | Gate to Source Charge | $V_{DD}$ = −15 V, $i_D$ = −22.1 A | – | 22 | – | |
| $Q_{gd}$ | Gate to Drain “Miller” Charge | $V_{DD}$ = −15 V, $i_D$ = −22.1 A | – | 46 | – | |
| **DRAIN−SOURCE DIODE CHARACTERISTICS** | | | | | | |
| $V_{SD}$ | Source to Drain Diode Forward Voltage | $V_{GS}$ = 0 V, $I_S$ = −2.1 A (Note 2) | – | 0.68 | 1.2 | V |
| | | $V_{GS}$ = 0 V, $I_S$ = −22.1 A (Note 2) | – | 0.79 | 1.25 | |
| $t_{rr}$ | Reverse Recovery Time | $I_F$ = −22.1 A, di/dt = 100 A/μs | – | 44 | 71 | ns |
| $Q_{rr}$ | Reverse Recovery Charge | | – | 39 | 63 | nC |

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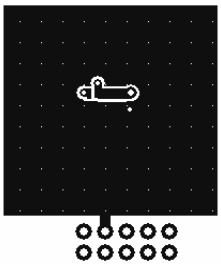
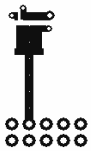
Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

NOTES:

1. $R_{\theta JA}$ is determined with the device mounted on a 1 in$^2$ pad 2 oz copper pad on a 1.5 × 1.5 in. board of FR−4 material. $R_{\theta JC}$ is guaranteed by design while $R_{\theta CA}$ is determined by the user’s board design.



a) 50°C/W when mounted on a 1 in$^2$ pad of 2 oz copper.



b) 125°C/W when mounted on a minimum pad of 2 oz copper.

2. Pulse Test: Pulse Width < 300 μs, Duty cycle < 2.0%.
3. The diode connected between the gate and source serves only as protection against ESD. No gate overvoltage rating is implied.
4. Pulsed $I_D$ please refer to Figure 12 SOA graph for more details.
5. Computed continuous current limited to Max Junction Temperature only, actual continuous current will be limited by thermal electro−mechanical application board design.

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## TYPICAL CHARACTERISTICS

($T_J$ = 25°C unless otherwise noted)

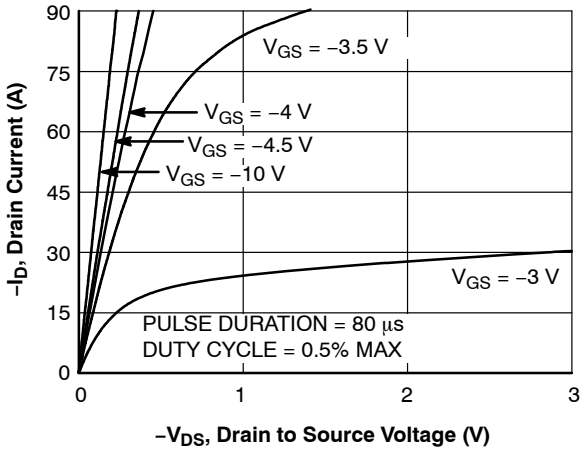


**Figure 1. On Region Characteristics**

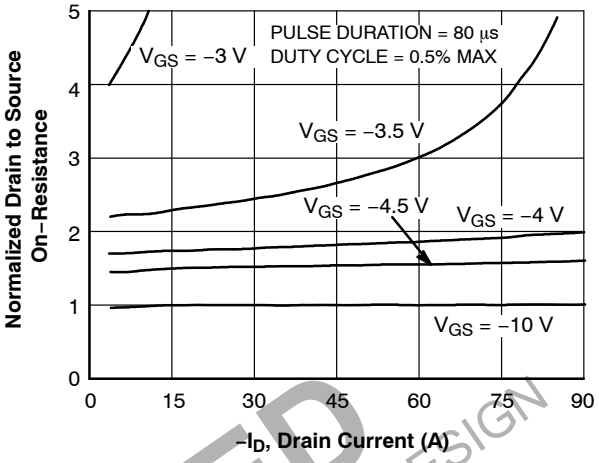


**Figure 2. Normalized On−Resistance vs. Drain Current and Gate Voltage**

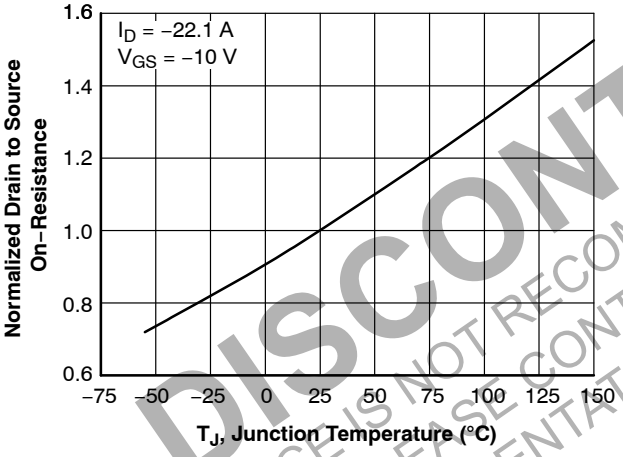


**Figure 3. Normalized On Resistance vs. Junction Temperature**

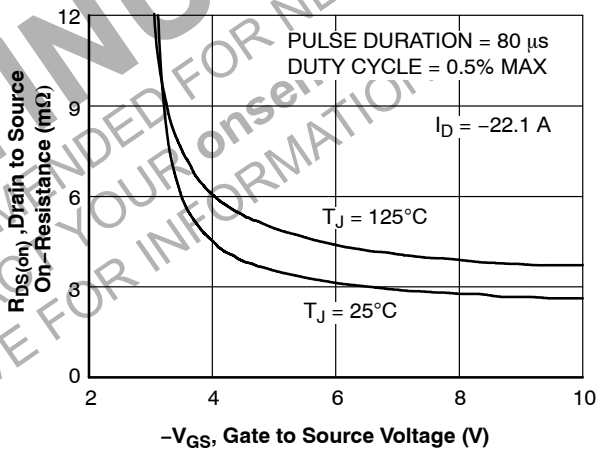


**Figure 4. On−Resistance vs. Gate to Source Voltage**

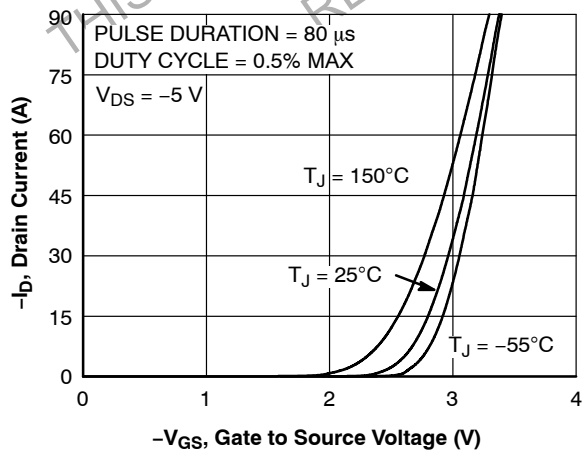


**Figure 5. Transfer Characteristics**

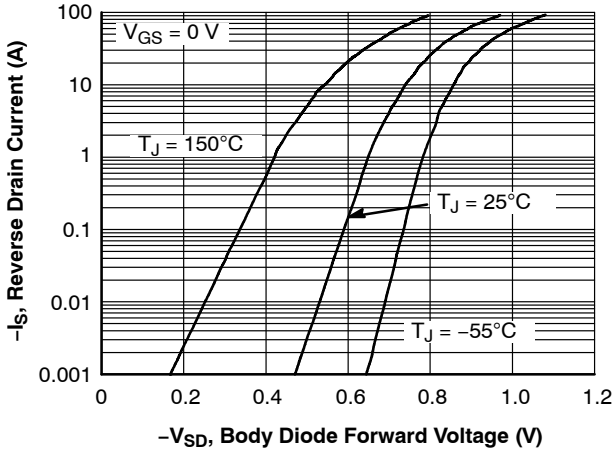


**Figure 6. Source to Drain Diode Forward Voltage vs. Source Current**

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## TYPICAL CHARACTERISTICS (continued)

($T_J$ = 25°C unless otherwise noted)

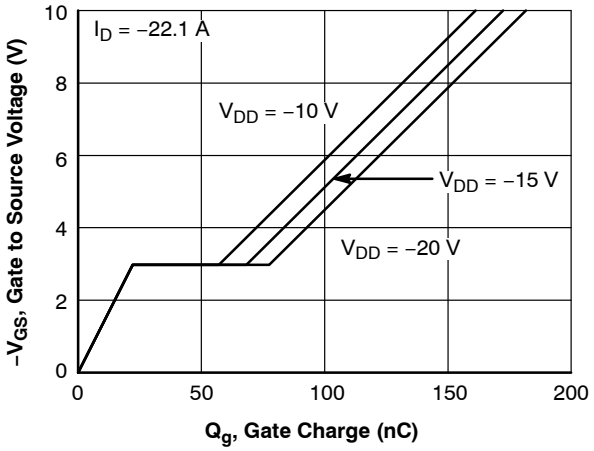


**Figure 7. Gate Charge Characteristics**

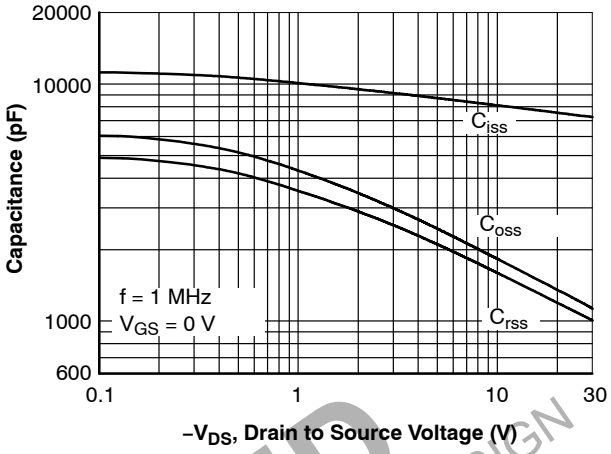


**Figure 8. Capacitance vs. Drain to Source Voltage**

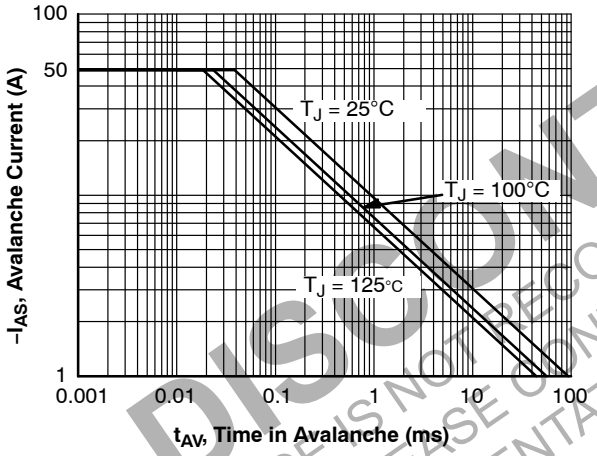


**Figure 9. Unclamped Inductive Switching Capability**

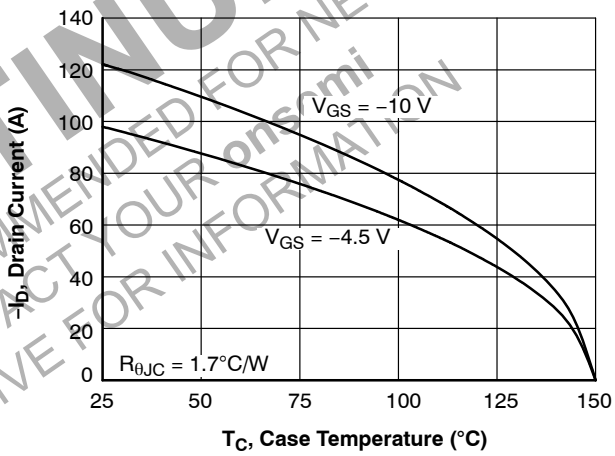


**Figure 10. Maximum Continuous Drain Current vs. Case Temperature**

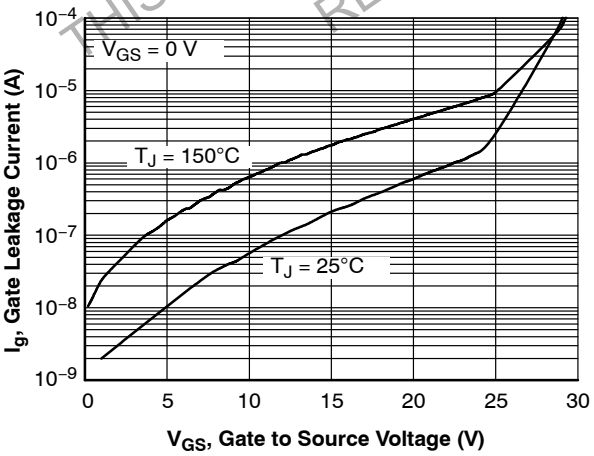


**Figure 11. $I_{gss}$ vs. $V_{gss}$**

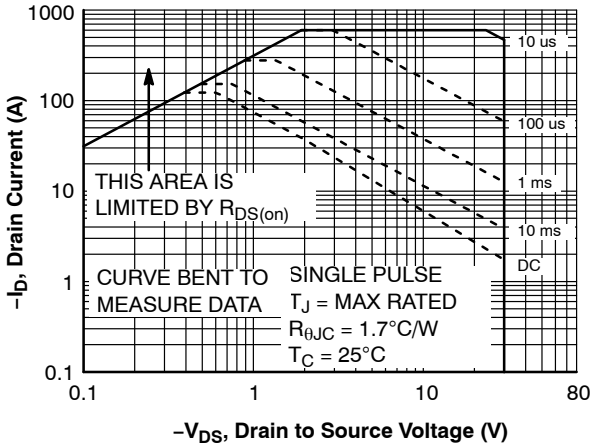


**Figure 12. Forward Bias Safe Operating Area**

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## TYPICAL CHARACTERISTICS (continued)

($T_J$ = 25°C unless otherwise noted)

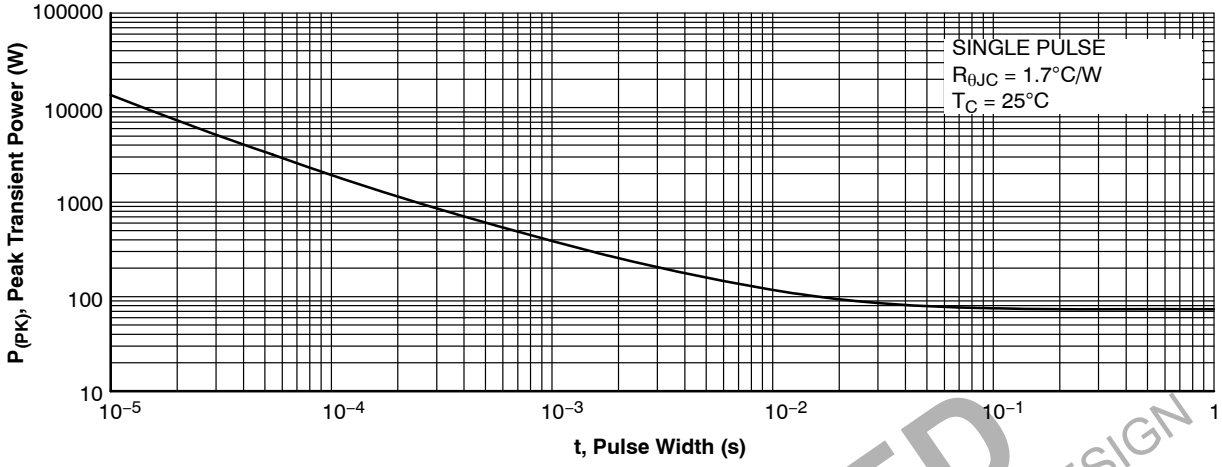


**Figure 13. Single Pulse Maximum Power Dissipation**

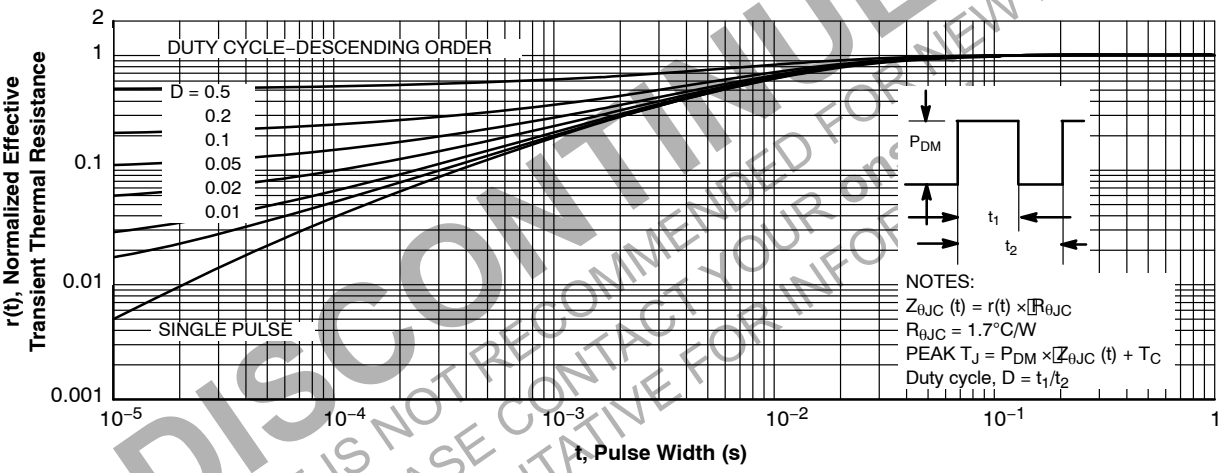


**Figure 14. Transient Thermal Response Curve**

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## PACKAGE MARKING AND ORDERING INFORMATION

| Device Marking | Device | Package | Shipping† |
|---|---|---|---|
| FDMS6681Z | FDMS6681Z | Power 56 | 3,000 Units/Tape & Reel |

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

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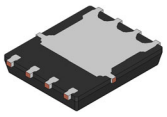
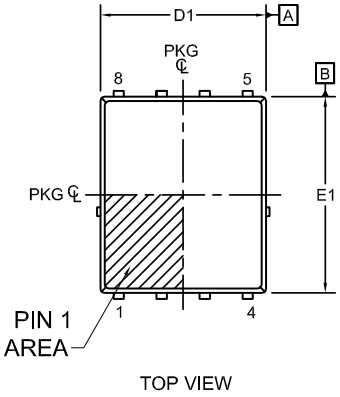
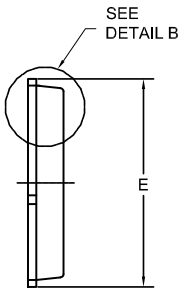


**PQFN8 5X6, 1.27P**
CASE 483AE
ISSUE C

DATE 21 JAN 2022



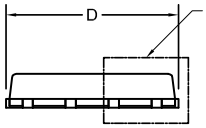

TOP VIEW



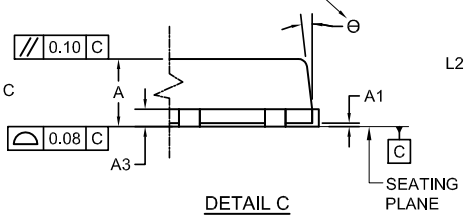

OPTIONAL DRAFT ANGLE MAY APPEAR ON FOUR SIDES OF THE PACKAGE

NOTES:

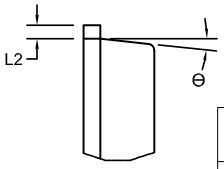
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 2009.
2. CONTROLLING DIMENSION: MILLIMETERS
3. COPLANARITY APPLIES TO THE EXPOSED PADS AS WELL AS THE TERMINALS.
4. DIMENSIONS D1 AND E1 DO NOT INCLUDE MOLD FLASH, PROTRUSIONS, OR GATE BURRS.
5. SEATING PLANE IS DEFINED BY THE TERMINALS. "A1" IS DEFINED AS THE DISTANCE FROM THE SEATING PLANE TO THE LOWEST POINT ON THE PACKAGE BODY.
6. IT IS RECOMMENDED TO HAVE NO TRACES OR VIAS WITHIN THE KEEP OUT AREA.




SIDE VIEW

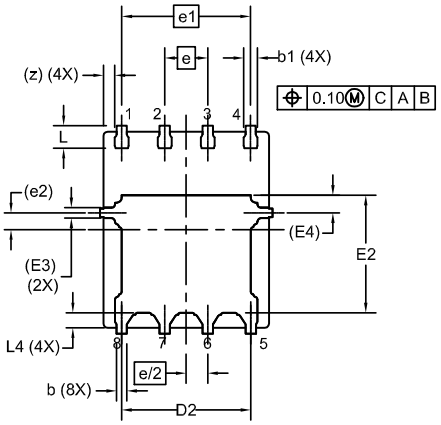



DETAIL C
SCALE: 2:1

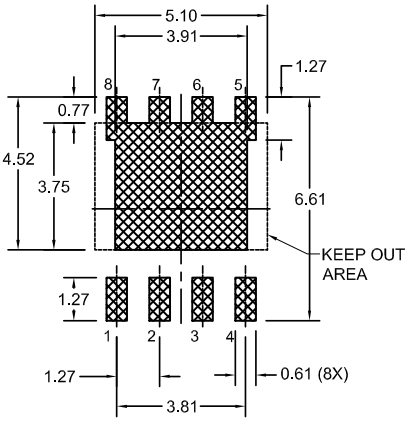



DETAIL B
SCALE: 2:1

| DIM | MILLIMETERS | | |
|---|---|---|---|
| | MIN. | NOM. | MAX. |
| A | 0.90 | 1.00 | 1.10 |
| A1 | 0.00 | - | 0.05 |
| b | 0.21 | 0.31 | 0.41 |
| b1 | 0.31 | 0.41 | 0.51 |
| A3 | 0.15 | 0.25 | 0.35 |
| D | 4.90 | 5.00 | 5.20 |
| D1 | 4.80 | 4.90 | 5.00 |
| D2 | 3.61 | 3.82 | 3.96 |
| E | 5.90 | 6.15 | 6.25 |
| E1 | 5.70 | 5.80 | 5.90 |
| E2 | 3.38 | 3.48 | 3.78 |
| E3 | 0.30 REF | | |
| E4 | 0.52 REF | | |
| e | 1.27 BSC | | |
| e/2 | 0.635 BSC | | |
| e1 | 3.81 BSC | | |
| e2 | 0.50 REF | | |
| L | 0.51 | 0.66 | 0.76 |
| L2 | 0.05 | 0.18 | 0.30 |
| L4 | 0.34 | 0.44 | 0.54 |
| z | 0.34 REF | | |
| Θ | 0° | - | 12° |




BOTTOM VIEW




LAND PATTERN RECOMMENDATION

*FOR ADDITIONAL INFORMATION ON OUR PB-FREE STRATEGY AND SOLDERING DETAILS, PLEASE DOWNLOAD THE ON SEMICONDUCTOR SOLDERING AND MOUNTING TECHNIQUES REFERENCE MANUAL, SOLDERRM/D.

| | | |
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| **DESCRIPTION:** | **PQFN8 5X6, 1.27P** | **PAGE 1 OF 1** |

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